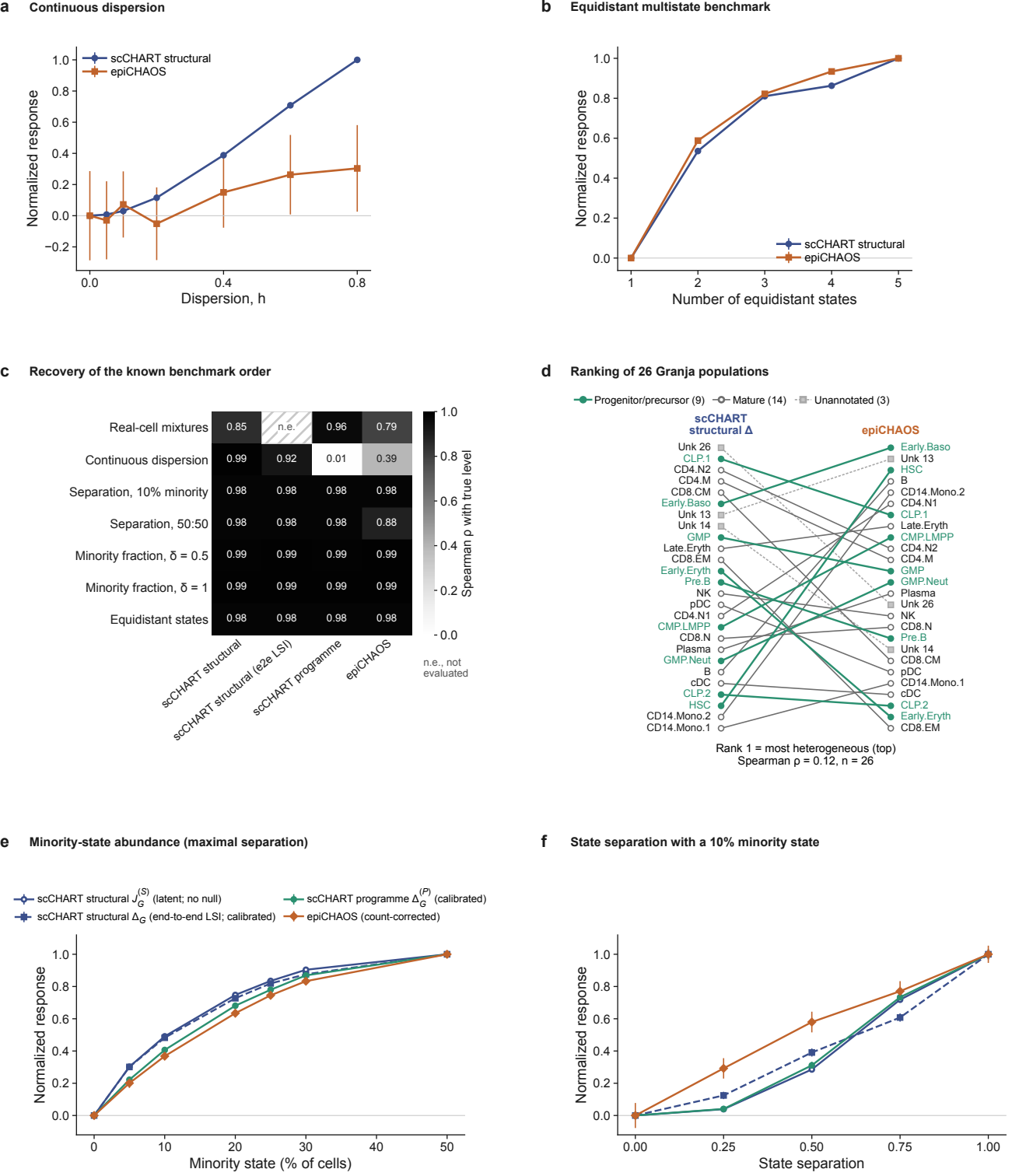

a
Continuous dispersion
scCHART structural
epiCHAOS
Normalized response
Dispersion, h
b
Equidistant multistate benchmark
Normalized response
Number of equidistant states
c
Recovery of the known benchmark order
Real-cell mixtures
Continuous dispersion
Separation, 10% minority
Separation, 50:50
Minority fraction, δ = 0.5
Minority fraction, δ = 1
Equidistant states
scCHART structural
scCHART structural (e2e LSI)
scCHART programme
epiCHAOS
Spearman ρ with true level
n.e., not evaluated
d
Ranking of 26 Granja populations
Progenitor/precursor (9)
Mature (14)
Unannotated (3)
scCHART structural Δ
epiCHAOS
Rank 1 = most heterogeneous (top)
Spearman ρ = 0.12, n = 26
e
Minority-state abundance (maximal separation)
scCHART structural $J_G^{(S)}$ (latent; no null)
scCHART structural Δ_G (end-to-end LSI; calibrated)
scCHART programme $\Delta_G^{(P)}$ (calibrated)
epiCHAOS (count-corrected)
Normalized response
Minority state (% of cells)
f
State separation with a 10% minority state
Normalized response
State separation